Modular Arithmetic

- $x \equiv 457 \pmod{13} = 2$

$$\begin{array}{r|l} 457 & 13 \\ \underline{39} & 35 \\ 67 & \\ \underline{65} & \\ 2 & \end{array}$$

- $x = -25 \pmod 7 = 3$

$$\begin{array}{r|l} -25 & 7 \\ \underline{-28} & -4 \\ 3 & \end{array}$$

The euclidean algorithm

- Calculate gcd (1160, 480)

$$1160 = 480 \times 2 + 200$$
$$480 = 200 \times 2 + 80$$
$$200 = 80 \times 2 + 40$$
$$80 = 40 \times 2 + 0$$

respuesta $= 40$

- $1160x + 480y = \gcd(1160, 480)$

$\gcd(1160, 480) = 40$

$40 = 200 - 80 \times 2$

Sustitución : $80 = 480 - 200(2)$

$$40 = 200 - (480 - 200(2))2$$
$$40 = 200 - 2(480 - (2)200)$$
$$40 = 200 - 2(480) - 4(200)$$
$$40 = 5(200) - 2(480)$$

Sustitución : $200 = 1160 - 480(2)$

$$40 = 5(1160 - 480(2)) - 2(480)$$
$$40 = 5(1160) - 10(480) - 2(480)$$
$$40 = 5(1160) - 12(480)$$

$x = 5, \quad y = -12$

Euler's Totient

- $\phi(21) = 21\left(1-\frac{1}{3}\right)\left(1-\frac{1}{7}\right)$

  $= 21 \cdot \frac{2}{3} \cdot \frac{6}{7}$

  $= 21 \cdot \frac{12}{21} = 12$

- $\phi(17) = 17 - 1$ 17 es numero primo

  $= 16$

- $\phi(360)$ using factorization

$$\begin{array}{r|l} 360 & 2 \\ 180 & 2 \\ 90 & 2 \\ 45 & 3 \\ 15 & 3 \\ 5 & 5 \end{array} \quad 2^3 \quad 3^2 \quad 5$$

$\phi(360) = 360\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right)$

$= 360\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{4}{5}\right)$

$= 96$

Information Theory

$P(A) = 1/2$, $P(B) = 1/4$, $P(C) = 1/8$, $P(D) = 1/8$

$I(x) = -\log_2 P(x)$

$I(A) = -\log_2(1/2) = 1$ bits
$I(B) = -\log_2(1/4) = 2$ bits
$I(C) = -\log_2(1/8) = 3$ bits
$I(D) = -\log_2(1/8) = 3$ bits

Entropy

$H(x) = -\sum P(x) \log_2 P(x)$

$= \frac{1}{2}(1) + \frac{1}{4}(2) + \left(\frac{1}{8}\right)(3) + \left(\frac{1}{8}\right)(3)$

$= 1.75$ bits

Bayes theorem:

$P(A)$ = Probability attack 0.001
$P(\neg A) = 0.999$
$P(F|A) = 0.99$ (true positive)
$P(F|\neg A) = 0.01$ (false positive)

$$P(F) = P(F|A)P(A) + P(F|\neg A)P(\neg A)$$

$$P(F) = (0.99)(0.001) + (0.01)(0.999)$$

$$P(F) = 0.00099 + 0.0099$$

$$P(F) = 0.01098$$

$$P(A|F) = \frac{0.99 \cdot 0.001}{0.01098}$$

$$= \frac{0.00099}{0.01098}$$

$$= 0.0902$$

Aunque el IDS tiene 99% de precisión, si señala un evento solo hay un 9% de posibilidades de que sea real